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United States Patent	12401119
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Toyao; Hiroshi et al.

Multiband antenna

Abstract

First and second slot edge portions of a conductive main portion are long in a first direction and sandwich a slot in a second direction. An open portion is formed a part different from the first slot edge portion and opens the slot outside the conductive main portion. A first part of a radiation element extends from an end portion of the first slot edge portion in the second direction. A second part of the radiation element extends from an end portion of the first part in the first direction. An additional element extends from the second part toward a second specific area through a first specific area. In a third direction, the first specific area and the second specific area overlap with the first slot edge portion and the second slot edge portion, respectively.

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Appl. No.:	18/127176
Filed:	March 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230335901 A1	Oct. 19, 2023

Foreign Application Priority Data

JP	2022-068160	Apr. 18, 2022
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Publication Classification

Int. Cl.: H01Q5/307 (20150101); H01Q1/38 (20060101); H01Q1/48 (20060101); H01Q5/378 (20150101); H01Q13/10 (20060101)

U.S. Cl.:

CPC H01Q5/307 (20150115); H01Q1/48 (20130101); H01Q13/106 (20130101);

Field of Classification Search

CPC: H01Q (1/243); H01Q (13/10); H01Q (1/48); H01Q (5/371); H01Q (1/38); H01Q (5/378); H01Q (9/0421); H01Q (9/42); H01Q (5/364); H01Q (13/106); H01Q (21/28); H01Q (5/35); H01Q (9/0407)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority under 35 U.S.C. § 119 to Japanese Patent Application No. JP2022-068160 filed Apr. 18, 2022, the contents of which are incorporated herein in their entirety by reference.

BACKGROUND OF THE INVENTION

(2) This invention relates to a multiband antenna, particularly, to a multiband antenna provided with a slot antenna and a radiation element.

(3) Referring to FIG. 17, a multiband antenna 90 disclosed in JP 2021-136527A (Patent Document

1) is provided with a slot antenna **92** and a radiation element **94**.

(4) As shown in FIG. **17**, a slot **921** of the slot antenna **92** has a longitudinal direction in a first direction or a Y-direction. The radiation element **94** has a first part **941** and a second part **943**. The first part **941** extends from the slot antenna **92** in a second direction or an X-direction perpendicular to the first direction. The second part **943** extends from an end portion of the first part **941** in the first direction. The second part **943** is larger than the first part **941** in length.

(5) The multiband antenna **90** of Patent Document 1 has two resonant frequencies or operating frequencies, namely, a resonant frequency of the slot antenna **92** and a resonant frequency of the radiation element **94**. Here, the second part **943** of the radiation element **94** extends in the first direction and lowers the resonant frequency of the slot antenna **92** in comparison with a case where the radiation element **94** is not provided. This means that the use of the radiation element **94** can cause downsizing of the slot antenna **90** which has a specific resonant frequency.

SUMMARY OF THE INVENTION

(6) It is an object of the present invention to provide a multiband antenna which can be downsized by adopting a structure different from that of the multiband antenna of Patent Document 1.

(7) One aspect of the present invention provides a multiband antenna which comprises a conductive main portion forming a slot antenna, a radiation element and an additional element. The conductive main portion comprises a first slot edge portion and a second slot edge portion. The conductive main portion is formed with a slot and an open portion. The slot has a longitudinal direction in a first direction. Each of the first slot edge portion and the second slot edge portion has a longitudinal direction in the first direction. The first slot edge portion and the second slot edge portion are arranged so that the first slot edge portion and the second slot edge portion sandwich the slot therebetween in a second direction perpendicular to the first direction. The open portion is formed at a part of the conductive main portion which is different from the first slot edge portion and opens the slot outside of the conductive main portion. The radiation element has a first part and a second part. The first part extends from an end portion of the first slot edge portion in the second direction. The second part extends from an end portion of the first part in the first direction. The additional element extends from the second part to or toward a second specific area through a first specific area without being brought into contact with the conductive main portion. The first specific area is an area which overlaps with the first slot edge portion in a third direction perpendicular to both the first direction and the second direction. The second specific area is an area which overlaps with the second slot edge portion in the third direction.

(8) In the multiband antenna of the present invention, the additional element adjusts an impedance of the multiband antenna, and thereby a resonant frequency of the slot antenna can be lowered. In other words, the additional element can downsize the slot antenna having a specific resonant frequency, so that the multiband antenna can be downsized.

(9) An appreciation of the objectives of the present invention and a more complete understanding of its structure may be had by studying the following description of the preferred embodiment and by referring to the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a perspective view showing a multiband antenna according to an embodiment of the present invention.

(2) FIG. **2** is a top view showing the multiband antenna of FIG. **1**. An additional element is depicted by a broken line in part. Each of a first specific area and a second specific area is depicted by a chain double-dashed line.

(3) FIG. **3** is a perspective view showing a first modification of the multiband antenna of FIG. **1**.

- (4) FIG. 4 is a perspective view showing a second modification of the multiband antenna of FIG. 1.
- (5) FIG. 5 is a perspective view showing a third modification of the multiband antenna of FIG. 1.
- (6) FIG. 6 is a perspective view showing a fourth modification of the multiband antenna of FIG. 1.
- (7) FIG. 7 is a perspective view showing a fifth modification of the multiband antenna of FIG. 1.
- (8) FIG. 8 is a perspective view showing a sixth modification of the multiband antenna of FIG. 1.
- (9) FIG. 9 is a perspective view showing a seventh modification of the multiband antenna of FIG. 1.
- (10) FIG. 10 is a perspective view showing an eighth modification of the multiband antenna of FIG. 1.
- (11) FIG. 11 is a perspective view showing a ninth modification of the multiband antenna of FIG. 1.
- (12) FIG. 12 is a perspective view showing a tenth modification of the multiband antenna of FIG. 1.
- (13) FIG. 13 is a perspective view showing an eleventh modification of the multiband antenna of FIG. 1. Conductive patterns formed on an object are depicted by broken lines.
- (14) FIG. 14 is a top view showing a twelfth modification of the multiband antenna of FIG. 1. An additional element is depicted by broken lines in part. Each of a first specific area and a second specific area is depicted by a chain double-dashed line.
- (15) FIG. 15 is a top view showing a thirteenth modification of the multiband antenna of FIG. 1. An additional element is depicted by broken lines in part. Each of a first specific area and a second specific area is depicted by a chain double-dashed line.
- (16) FIG. 16 is a bottom view showing a fourteenth modification of the multiband antenna of FIG. 1. Each of a first specific area and a second specific area is depicted by a chain double-dashed line.
- (17) FIG. 17 is a top view showing a multiband antenna disclosed in Patent Document 1.
- (18) While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof are shown by way of example in the drawings and will herein be described in detail. It should be understood, however, that the drawings and detailed description thereto are not intended to limit the invention to the particular form disclosed, but on the contrary, the intention is to cover all modifications, equivalents and alternatives falling within the spirit and scope of the present invention as defined by the appended claims.

DETAILED DESCRIPTION

- (19) Referring to FIG. 1, a multiband antenna **10** according to an embodiment of the present invention is provided with a conductive main portion **20**, a first radiation element (radiation element) **30** and an additional element **40**. In the present embodiment, the multiband antenna **10** is further provided with a second radiation element (additional radiation element) **50** and a grounding terminal **60**. However, in the present invention, the second radiation element **50** and the grounding terminal **60** are not essential. Nevertheless, by providing the second radiation element **50**, a bandwidth of the multiband antenna **10** can be widened.
- (20) As understood from FIG. 1, the multiband antenna **10** is formed of a single sheet metal. In other words, the multiband antenna **10** is formed by punching and bending a single metal sheet. However, the present invention is not limited thereto. The multiband antenna **10** may be formed of a plurality of metal sheets. Alternatively, the multiband antenna **10** may be formed of a metal foil or a conductive pattern formed on a circuit board instead of the metal sheet at least in part. Furthermore, the multiband antenna **10** may be formed of a metal sheet or a metal foil and a supporter which is made of resin and supports the metal sheet or the metal foil if necessary.
- (21) As shown in FIG. 1, the conductive main portion **20** has at least a first slot edge portion **201** and a second slot edge portion **203**. In the present embodiment, the conductive main portion **20** further has a coupling portion **205**. The first slot edge portion **201**, the second slot edge portion **203** and the coupling portion **205** are coupled to one another and define a slot **22** and an open portion **24**. In other words, the conductive main portion **20** is formed with the slot **22** and the open portion

24.

(22) As understood from FIG. 1, the conductive main portion **20** is located on a specific plane defined by a first direction and a second direction perpendicular to the first direction. In the present embodiment, the first direction is a Y-direction, and the second direction is an X-direction. Moreover, in the present embodiment, the specific plane is an X-Y plane. The first direction defines a first orientation and a second orientation which is an orientation opposite to the first orientation. In the present embodiment, the first orientation is a negative Y-direction, and the second orientation is a positive Y-direction. Moreover, in the present embodiment, the second direction is also a front-rear direction. A negative X-direction is directed forward while a positive X-direction is directed rearward.

(23) As shown in FIG. 1, each of the first slot edge portion **201** and the second slot edge portion **203** has a rectangular shape long in a first direction. The first slot edge portion **201** has an end portion **2011**, which is oriented in the first orientation of the first direction, and an end portion **2013**, which is oriented in the second orientation of the first direction. The second slot edge portion **203** has an end portion **2031**, which is oriented in the first orientation of the first direction, and an end portion **2033**, which is oriented in the second orientation of the first direction.

(24) As shown in FIG. 1, the first slot edge portion **201** and the second slot edge portion **203** are positioned apart from each other in the second direction. The first slot edge portion **201** is located forward of the second slot edge portion **203**. In the second direction, the slot **22** and the open portion **24** are located between the first slot edge portion **201** and the second slot edge portion **203**. Thus, the first slot edge portion **201** and the second slot edge portion **203** are arranged so that they sandwich the slot **22** and the open portion **24** therebetween.

(25) As shown in FIG. 1, the coupling portion **205** has a rectangular shape long in the second direction. The coupling portion **205** couples one of the end portions of the first slot edge portion **201** to one of the end portions of the second slot edge portion **203**. In the present embodiment, the coupling portion **205** couples the end portion **2013** of the first slot edge portion **201** and the end portion **2033** of the second slot edge portion **203** to each other.

(26) As shown in FIG. 1, the slot **22** has a longitudinal direction in the first direction. The open portion **24** is located at an end portion of the conductive main portion **20**, wherein the end portion of the conductive main portion **20** is oriented in the first orientation. In other words, the open portion **24** is located between the end portion **2011** of the first slot edge portion **201** and the end portion **2031** of the second slot edge portion **203**. The open portion **24** is contiguous to the slot **22** and opens the slot **22** outside of the conductive main portion **20**. In the present embodiment, the open portion **24** is opened in the first orientation of the first direction. However, the present invention is not limited thereto. The open portion **24** may be opened forward or rearward. Even when the open portion **24** is opened forward or rearward, the open portion **24** of the present invention is formed at a part of the conductive main portion which is different from the first slot edge portion **201**.

(27) As shown in FIG. 1, the first radiation element **30** has a first part **301** and a second part **303**. The first part **301** has a rectangular shape long in the second direction and is located on a specific plane. The first part **301** extends in the second direction from the end portion **2011** of the first slot edge portion **201**, wherein the end portion **2011** is oriented in the first orientation of the first direction. In the present embodiment, the first part **301** extends rearward.

(28) As shown in FIG. 1, the second part **303** of the first radiation element **30** extends from a rear end portion of the first part **301** in the second orientation of the first direction. In the present embodiment, the second part **303** has an upper portion **321** and a rear portion **323**. The upper portion **321** has a rectangular shape long in the first direction and is located on the specific plane. The rear portion **323** has a rectangular shape long in the first direction and extends from a rear edge of the upper portion **321** in a third direction perpendicular to both the first direction and the second direction. In the present invention, the rear portion **323** is not essential. However, the rear portion

323 can increase a radiation efficiency of the first radiation element **30** without increasing an occupation area of the first radiation element **30** when viewed along the third direction. In the present embodiment, the third direction is a Z-direction. Supposing a positive Z-direction is directed upward while a negative Z-direction is directed downward, the rear portion **323** extends downward from the upper portion **321**.

(29) As shown in FIG. 1, the additional element **40** extends forward from a lower edge of the rear portion **323** of the second part **303** of the first radiation element **30**. The additional element **40** is positioned apart from the conductive main portion **20** in the third direction and extends forward without being brought into contact with the conductive main portion **20**.

(30) Referring to FIG. 2, in the present embodiment, the additional element **40** has a rectangular shape. The additional element **40** is positioned apart from both ends of the second part **303** in the first direction. Moreover, the additional element **40** is nearer to an end portion of the second part **303**, which is oriented in the second orientation, than to an end portion of the second part **303**, which is oriented in the first orientation, in the first direction. However, the present invention is not limited thereto. The shape and the position of the additional element **40** may be freely set according to intended antenna properties.

(31) As understood from FIG. 2, the additional element **40** extends toward a second specific area **75** through a first specific area **70**. In the present embodiment, the additional element **40** extends to the second specific area **75**. In other words, the additional element **40** overlaps with the second slot edge portion **203** when viewed along the third direction. Here, each of the first specific area **70** and the second specific area **75** is an area on a plane which is perpendicular to the third direction and which is positioned apart from the specific plane in the third direction. In addition, the first specific area **70** is an area overlapping with the first slot edge portion **201** in the third direction. Moreover, the second specific area **75** is an area overlapping with the second slot edge portion **203** in the third direction. In the present embodiment, the additional element **40** is located on a plane in which the first specific area **70** and the second specific area **75** are included, and a front edge **401** of the additional element **40** is in the second specific area **75**. However, the present invention is not limited thereto. Each of the first specific area **70** and the second specific area **75** may be freely set according to intended antenna properties.

(32) As understood from FIGS. 1 and 2, the front edge **401** of the additional element **40** is located near to the second slot edge portion **203**. With this structure, a capacitor is formed between the additional element **40** and the second slot edge portion **203**. By setting a shape and a size of the additional element **40** to give an intended value to a capacitance, an impedance of the multiband antenna **10** can be adjusted, and downsizing of the multiband antenna **10** can be achieved. Although the additional element **40** extends to the second specific area **75** in the present embodiment, the additional element **40** may not extend to the second specific area **75**. However, if an area where the additional element **40** and the second slot edge portion **203** overlap with each other is larger when viewed along the third direction, larger capacitance can be obtained. Larger capacitance can achieve a lower operating frequency and downsize the multiband antenna **10**.

(33) As shown in FIGS. 1 and 2, the second radiation element **50** is located on the specific plane and extends from the first radiation element **30** in the first orientation. In detail, the second radiation element **50** has a long portion **501** and a short portion **503**. The long portion **501** has a rectangular shape long in the first direction. Moreover, the short portion **503** has a rectangular shape long in the second direction. The long portion **501** extends in the first orientation from the end portion of the second part **303** of the first radiation element **30**, wherein the end portion of the second part **303** is oriented in the first orientation of the first direction. The short portion **503** extends forward from an end portion of the long portion **501**, wherein the end portion of the long portion **501** is oriented in the first orientation of the first direction. However, the present invention is not limited thereto. The second radiation element **50** may be formed of only the long portion **501**. However, the short portion **503** can elongate an electrical length of the second radiation element **50**.

without increasing a size of the second radiation element **50** in the first direction.

(34) As shown in FIG. 2, the grounding terminal **60** has a rectangular shape long in the second direction. The grounding terminal **60** extends forward from a front edge of the second slot edge portion **203**. In detail, the grounding terminal **60** extends forward from a front edge of the end portion **2031** of the second slot edge portion **203**. An edge of the grounding terminal **60**, which is oriented in the first orientation of the first direction, is arranged on a straight line with an edge of the second slot edge portion **203**, which is oriented in the first orientation of the first direction. However, the present invention is not limited thereto. The shape, the size and the position of the grounding terminal **60** may be freely set according to intended properties.

(35) The grounding terminal **60** is connected to a host conductor (not shown) when used. The host conductor may be a device case (not shown) which accommodates the multiband antenna **10** or a ground pattern of a circuit board (not shown) on which the multiband antenna **10** is mounted. By using the host conductor, downsizing of the multiband antenna **10** can be achieved.

(36) As shown in FIGS. 1 and 2, the conductive main portion **20** is provided with feeding points **211** and **213**. In the present embodiment, the feeding points **211** and **213** are located nearer to the coupling portion **205** than to the open portion **24** in the first direction. The feeding points **211** and **213** are located so that they sandwich the slot **22** in the second direction. By supplying high-frequency power between the feeding points **211** and **213**, the multiband antenna **10** is operated as an antenna. For supplying the high-frequency power between the feeding points **211** and **213**, a coaxial cable (not shown) may be used, for example.

(37) The multiband antenna **10** has a plurality of operating frequencies. In detail, the multiband antenna **10** has three operating frequencies depending on the conductive main portion **20**, the first radiation element **30** and the second radiation element **50**, respectively. An electrical length of each of the first radiation element **30** and the second radiation element **50** is equal to a quarter of a wavelength of the operating frequency corresponding thereto. The electrical length of the first radiation element **30** and the electrical length of the second radiation element **50** are different from each other. For example, the electrical length of the second radiation element **50** is longer than the electrical length of the first radiation element **30**. With this structure, the second radiation element **50** can have the operating frequency lower than that of the first radiation element **30**. The operating frequency depending on the conductive main portion **20** is lower than that of only the conductive main portion **20** because of influence of each of the first radiation element **30**, the second radiation element **50** and the grounding terminal **60**. Accordingly, when trying to obtain a specific operating frequency, each of the first radiation element **30**, the second radiation element **50** and the grounding terminal **60** helps to downsize the multiband antenna **10**. The additional element **40** adjusts the impedance of the multiband antenna **10** and lowers the operating frequencies of the multiband antenna **10** or helps to downsize the multiband antenna **10**.

(38) Although the description about one embodiment of the present invention is made above, the multiband antenna **10** may be modified as follows. In each of modifications mentioned below, the same or the similar components same as or similar to those of the multiband antenna **10** are represented by the same or the similar reference signs and the description thereabout is omitted.

(39) [First Modification]

(40) Referring to FIG. 3, a multiband antenna **10A** of a first modification is different from the multiband antenna **10** (see FIG. 1) of the aforementioned embodiment in that positions of feeding points **211A** and **213A** are different from those of the feeding points **211** and **213** (see FIG. 1).

(41) As shown in FIG. 3, the positions of the feeding points **211A** and **213A** are nearer to an open portion **24** than to a coupling portion **205** in the first direction. Thus, in the present invention, the positions of the feeding points **211** and **213** or **211A** and **213A** may be changed according to intended antenna properties.

(42) [Second Modification]

(43) Referring to FIG. 4, a multiband antenna **10B** of a second modification is different from the

multiband antenna **10A** (see FIG. 3) of the first modification in that a shape of an additional element **40B** is different from that of the additional element **40** (see FIG. 3).

(44) As shown in FIG. 4, the additional element **40B** has an L-shape when viewed along the third direction. A size of a front edge **401B** of the additional element **40B** is larger than that of the front edge **401** of the additional element **40** in the first direction. With this structure, a capacitance between the additional element **40B** and a second slot edge portion **203** can be larger than that between the additional element **40** and the second slot edge portion **203**. A larger capacitance can achieve a lower operating frequency and downsize the multiband antenna **10B**.

(45) [Third Modification]

(46) Referring to FIG. 5, a multiband antenna **10C** of a third modification is different from the multiband antenna **10** (see FIG. 1) of the aforementioned embodiment in that it has a grounding terminal **60C** which has a part extending in a direction intersecting with the specific plane or the X-Y plane.

(47) As shown in FIG. 5, in the present modification, the grounding terminal **60C** has a rectangular flat plate-like shape, and the whole thereof extends downward from a front edge of a second slot edge portion **203**. However, the present invention is not limited thereto. The grounding terminal **60C** may extend forward from the front edge of the second slot edge portion **203** and then extend the direction intersecting with the specific plane. In that case, the part extending in the direction intersecting with the specific plane may be on a plane perpendicular to the first direction or on a plane perpendicular to the second direction.

(48) [Fourth Modification]

(49) Referring to FIG. 6, a multiband antenna **10D** of a fourth modification is different from the multiband antenna **10C** (see FIG. 5) of the third modification in that a position of a grounding terminal **60D** is different from that of the grounding terminal **60C** (see FIG. 5).

(50) As shown in FIG. 6, in the present modification, the grounding terminal **60D** is positioned apart from both ends of a second slot edge portion **203** in the first direction. Moreover, the grounding terminal **60D** is nearer to an open portion **24** than to a coupling portion **205** in the first direction. Thus, in the present invention, the position of the grounding terminal **60**, **60C** or **60D** may be changed according to intended antenna properties.

(51) [Fifth Modification]

(52) Referring to FIG. 7, a multiband antenna **10E** of a fifth modification is different from the multiband antenna **10C** (see FIG. 5) of the third modification in that it has an additional grounding terminal **60E** in addition to a grounding terminal **60C**.

(53) As shown in FIG. 7, in the present modification, the additional grounding terminal **60E** extends downward from a front edge of an end portion **2033** of a second slot edge portion **203**. The additional grounding terminal **60E** helps to improve reliability of the multiband antenna **10E**. Thus, the multiband antenna of the present invention can be provided with any number of grounding terminals.

(54) [Sixth Modification]

(55) Referring to FIG. 8, a multiband antenna **10F** of a sixth modification is different from the multiband antenna **10C** (see FIG. 5) of the third modification in that a shape of a second radiation element **50F** is different from that of the second radiation element **50** (see FIG. 5). In detail, in the multiband antenna **10F**, the second radiation element **50F** has an extension portion **505** in addition to a long portion **501** and a short portion **503**.

(56) As shown in FIG. 8, the extension portion **505** extends from a front-end portion of the short portion **503** in the second orientation. The extension portion **505** can lengthen an electrical length of the second radiation element **50F** without increasing a size of the second radiation element **50F** in the first direction. Thus, in the present invention, a shape of the second radiation element **50** or **50F** may be changed according to intended antenna properties.

(57) [Seventh Modification]

(58) Referring to FIG. 9, a multiband antenna **10G** of a seventh modification is different from the multiband antenna **10F** (see FIG. 8) of the sixth modification in that a shape of a second radiation element **50G** is different from that of the second radiation element **50F** (see FIG. 8). In detail, in the multiband antenna **10G**, the second radiation element **50G** has a vertical portion **507** in addition to the structure of the second radiation element **50F**.

(59) As shown in FIG. 9, the vertical portion **507** extends downward from a rear edge of a long portion **501**. In the first direction, a size of the vertical portion **507** is smaller than that of the long portion **501**. The vertical portion **507** helps to improve strength and radiation properties of the second radiation element **50G**. Thus, in the present invention, a shape of the second radiation element **50**, **50F** or **50G** may be changed according to intended antenna properties.

(60) [Eighth Modification]

(61) Referring to FIG. 10, a multiband antenna **10H** of an eighth modification is different from the multiband antenna **10G** (see FIG. 9) of the seventh modification in that it is provided with a third radiation element **53**.

(62) As shown in FIG. 10, the third radiation element **53** has an additional long portion **531**, an additional short portion **533** and an additional extension portion **535**. The third radiation element **53** is formed so that it is substantially same as a second radiation element **50G**. The additional long portion **531** is coupled with a lower edge of a vertical portion **507**. When viewed along the third direction, the third radiation element **53** overlaps with the second radiation element **50G**. Thus, in the multiband antenna of the present invention, the number of radiation elements or passive antennas, i.e., the number of operating frequencies or an operating frequency band can be freely set.

(63) [Ninth Modification]

(64) Referring to FIG. 11, a multiband antenna **10I** of a ninth modification is different from the multiband antenna **10C** (see FIG. 5) of the third modification in that it is provided with a fourth radiation element **55**.

(65) As shown in FIG. 11, the fourth radiation element **55** has a rectangular shape long in the first direction. The fourth radiation element **55** extends from an end portion **2011** of a first slot edge portion **201** in the first orientation. In the first direction, a size of the fourth radiation element **55** is equal to or less than half of a size of a long portion **501** of a second radiation element **50**. However, the present invention is not limited thereto. The shape and the size of the fourth radiation element **55** may be freely set according to intended antenna properties.

(66) [Tenth Modification]

(67) Referring to FIG. 12, a multiband antenna **10J** of a tenth modification is different from the multiband antenna **10C** (see FIG. 5) of the third modification in that it is provided with a fifth radiation element **57**.

(68) As shown in FIG. 12, the fifth radiation element **57** has a rectangular shape long in the first direction. The fifth radiation element **57** extends from an end portion **2031** of a second slot edge portion **203** in the first orientation. In the first direction, a size of the fifth radiation element **57** is equal to or less than half of a size of a long portion **501** of a second radiation element **50**. However, the present invention is not limited thereto. The shape and the size of the fifth radiation element **57** may be freely set according to intended antenna properties.

(69) [Eleventh Modification]

(70) Referring to FIG. 13, a multiband antenna **10K** of an eleventh modification is different from the multiband antenna **10E** (see FIG. 7) of the fifth modification in that it is provided with a feeding terminal **62**.

(71) As shown in FIG. 13, the feeding terminal **62** has a part extending in a direction intersecting with the specific plane. In the present modification, the feeding terminal **62** has a rectangular flat plate-like shape, and the whole thereof extends downward from a front edge of a first slot edge portion **201**. However, the present invention is not limited thereto. The feeding terminal **62** may

extend forward from the front edge of the first slot edge portion **201** and then extend the direction intersecting with the specific plane. In that case, the part extending in the direction intersecting with the specific plane may be on a plane perpendicular to the first direction or on a plane perpendicular to the second direction.

(72) As understood from FIG. **13**, in the third direction, a size of the feeding terminal **62** is equal to that of a grounding terminal **60C** and to that of an additional grounding terminal **60E**. With this structure, the multiband antenna **10K** can be surface mounted on an object (not shown), such as a circuit board. For example, if conductive patterns **80** corresponding to the feeding terminal **62**, the grounding terminal **60C** and the additional grounding terminal **60E**, respectively, are formed on the object, the feeding terminal **62**, the grounding terminal **60C** and the additional grounding terminal **60E** can be connected to the conductive patterns **80** corresponding to them, respectively.

(73) [Twelfth Modification]

(74) Referring to FIG. **14**, a multiband antenna **10L** of a twelfth modification is different from the multiband antenna **10** (see FIG. **2**) of the aforementioned embodiment in that it further has a first extension slot edge portion **207**.

(75) As shown in FIG. **14**, the first extension slot edge portion **207** has an L-shape when viewed along the third direction. In detail, the first extension slot edge portion **207** extends from an end portion **2011** of a first slot edge portion **201** in the first orientation and then extends forward. In the present modification, an open portion **24L** is formed at a part of a conductive main portion **20** which is different from the first slot edge portion **201**. In detail, the open portion **24L** is located between a front edge of a second slot edge portion **203** and a front edge of the first extension slot edge portion **207** and opened forward.

(76) [Thirteenth Modification]

(77) Referring to FIG. **15**, a multiband antenna **10M** of a thirteenth modification is different from the multiband antenna **10** (see FIG. **2**) of the aforementioned embodiment in that it further has a first extension slot edge portion **207M** and a second extension slot edge portion **209**.

(78) As shown in FIG. **15**, the first extension slot edge portion **207M** has a rectangular shape and extends from an end portion **2011** of a first slot edge portion **201** in the first orientation. Moreover, the second extension slot edge portion **209** has an inverted L-shape when viewed along the third direction. In detail, the second extension slot edge portion **209** extends from an end portion **2031** of a second slot edge portion **203** in the first orientation and then extends rearward. In the present modification, an open portion **24M** is formed at a part of a conductive main portion **20** which is different from the first slot edge portion **201**. In detail, the open portion **24M** is located between a rear edge of the first extension slot edge portion **207M** and a rear edge of the second extension slot edge portion **209** and opened rearward.

(79) [Fourteenth Modification]

(80) Referring to FIG. **16**, a multiband antenna **10N** of a fourteenth modification is different from the multiband antenna **10** (see FIG. **2**) of the aforementioned embodiment in that a shape of an additional element **40N** is different from that of the additional element **40** (see FIG. **2**). However, the additional element **40N** of the present modification has in common with the additional element **40** in that it extends toward a second specific area **75** through a first specific area **70**.

(81) In detail, as shown in FIG. **16**, the additional element **40N** of the present modification has a crank shape when viewed along the third direction. In more detail, the additional element **40N** of the present modification extends forward from a lower end of a rear portion **323** of a first radiation element **30** and then extends in the first orientation and further extends forward. Additionally, each of the first specific area **70** and the second specific area **75** is an area on a plane which is perpendicular to the third direction and which is positioned apart from the specific plane in the third direction. In addition, the first specific area **70** is an area overlapping with a first slot edge portion **201** in the third direction. Moreover, the second specific area **75** is an area overlapping with a second slot edge portion **203** in the third direction. In the present modification, the additional

element **40N** is located on a plane where the first specific area **70** and the second specific area **75** are included, and a front edge **401N** of the additional element **40N** is in the second specific area **75**. The additional element **40N** forms a capacitance between itself and the second slot edge portion **203** and adjusts an impedance of the multiband antenna **10N**, so that it lowers operating frequencies of the multiband antenna **10N** or helps to downsize the multiband antenna **10N**.

(82) Although the specific explanation about the present invention is made above with reference to concrete embodiments, the present invention is not limited thereto but susceptible of various modifications and alternative forms without departing from the spirit of the invention. For example, the structures of the modifications **1** to **14** may be suitably selected and combined.

(83) While there has been described what is believed to be the preferred embodiment of the invention, those skilled in the art will recognize that other and further modifications may be made thereto without departing from the spirit of the invention, and it is intended to claim all such embodiments that fall within the true scope of the invention.

Claims

1. A multiband antenna comprising a conductive main portion forming a slot antenna, a radiation element and an additional element, wherein: the conductive main portion comprises a first slot edge portion and a second slot edge portion; the conductive main portion is formed with a slot and an open portion; the slot has a longitudinal direction in a first direction; each of the first slot edge portion and the second slot edge portion has a longitudinal direction in the first direction; the first slot edge portion and the second slot edge portion are arranged so that the first slot edge portion and the second slot edge portion sandwich the slot therebetween in a second direction perpendicular to the first direction; the open portion is formed at a part of the conductive main portion which is different from the first slot edge portion and opens the slot outside of the conductive main portion; the radiation element has a first part and a second part; the first part extends from an end portion of the first slot edge portion in the second direction; the second part extends from an end portion of the first part in the first direction; the additional element extends from the second part to or toward a second specific area through a first specific area without being brought into contact with the conductive main portion; the first specific area is an area which overlaps with the first slot edge portion in a third direction perpendicular to both the first direction and the second direction; and the second specific area is an area which overlaps with the second slot edge portion in the third direction.
2. The multiband antenna as recited in claim 1, wherein the additional element extends to the second specific area and overlaps with the second slot edge portion when viewed along the third direction.
3. The multiband antenna as recited in claim 2, wherein: the multiband antenna further comprises an additional radiation element; the first part extends, in the second direction, from the end portion of the first slot edge portion which is directed in a first orientation of the first direction; the second part extends from the end portion of the first part in a second orientation opposite to the first orientation; and the additional radiation element extends from the radiation element in the first orientation.
4. The multiband antenna as recited in claim 3, wherein the additional radiation element extends from the second part of the radiation element.
5. The multiband antenna as recited in claim 1, wherein: the multiband antenna further comprises a grounding terminal; and the grounding terminal extends from the second slot edge portion.
6. The multiband antenna as recited in claim 5, wherein: the grounding terminal has a part extending in a direction intersecting with a specific plane which is defined by the first direction and

the second direction; the multiband antenna further comprises a feeding terminal; and the feeding terminal is a part extending in a direction intersecting with the specific plane.
